

Measurement Theory*

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Enrique Amigó¹ and Stefano Mizzaro²

¹UNED, Madrid, Spain, enrique@lsi.uned.es

²University of Udine, Udine, Italy, mizzaro@uniud.it

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In this appendix, we recall some basic concepts in measurement theory. Measurement theory is the discipline that studies the theoretical foundations of the activity of measuring. It is not a new field, and its roots are usually dated back to the seminal work by Stevens [3]. We try to make this paper self-contained: we therefore sacrifice some of the most technical issues when not necessary to our aims, as well as abuse the notation sometimes, and we provide some examples to help the intuition. A more formal and complete account can be found in the classical works by Suppes and Zinnes [4], Roberts [2], or Pedhazur and Schmelkin [1].

1 Assignments and Measurements

In measurement theory a measurement is defined as a function that assigns real numbers to elements of a set $\mathcal{X}$ of objects or events in the real world. To be a measurement, the function must be a homomorphism, i.e., it must be such that the relationships in the so called *empirical relational structure*¹ (i.e., the real world) are preserved in the *numerical relational structure* (i.e., the codomain of the function, the set of real numbers $\mathbb{R}$). For example, when measuring the weight of objects, we can have in the real world a relationship R that stands for “heavier than” and an operation $\circ$ of “combination” (considering two objects together). A measurement will assign numbers such that R corresponds to “ $>$ ” and $\circ$ to “ $+$ ”, thus defining a homomorphism from the empirical relational structure (the tuple $\langle \mathcal{X}, R, \circ \rangle$) to the numerical relational structure $(\langle \mathbb{R}, >, + \rangle)$.

For our purposes, we can start from a simpler situation: we define an assignment as a function that assigns a real number to a characteristic of an object or event.

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¹In this paper we prefer to use the term “structure” in place of the original “system” to avoid confusion, since the latter is already used for “IR system” and “system output”.

Definition 1 (Assignment) An assignment $\omega \in \Omega$ (we denote with Ω the set of all possible assignments) is a function that assigns values to objects in a set $\mathcal{D}$:

$$\omega : \mathcal{D} \longrightarrow \mathbb{R}.$$

Then, to be a measurement, an assignment must be a homomorphism [1, 2, 4].

Definition 2 (Measurement) Let $\langle \mathcal{D}, R_{\mathcal{D}} \rangle$ be an empirical relational structure and $\langle \mathbb{R}, R_{\mathbb{R}} \rangle$ be a numerical relational structure.² A measurement $\psi \in \Psi$ (we denote with $\Psi \subset \Omega$ the set of all possible measurements) is an assignment of objects in a set $\mathcal{D}$

$$\psi : \mathcal{D} \longrightarrow \mathbb{R}$$

that is a homomorphism between the empirical relational structure and the numerical relational structure, i.e., $\forall x, y \in \mathcal{D} (R_{\mathcal{D}}(x, y) \implies R_{\mathbb{R}}(\psi(x), \psi(y)))$.

This definition is similar to the classical ones that can be found in the literature [2, pp. 51–52] [4, pp. 4–8] [1, page 17]; it is very general and includes different kinds of measurements. Let us see some examples to emphasize this fact.

Example 1 Let us consider three objects $\mathcal{D} = \{o_1, o_2, o_3\}$, and the measurement ψ_t of their temperature in Celsius degrees; then $\psi_t = \{\langle o_1, -5 \rangle, \langle o_2, 5 \rangle, \langle o_3, 15 \rangle\}$ (we represent the function ψ_t as a set of pairs).

Example 2 Let us consider three products $\mathcal{D} = \{p_1, p_2, p_3\}$, and the quality measurement ψ_q into high (we use the real number 2 for this), medium (1) or low (0); then $\psi_q = \{\langle p_1, 0 \rangle, \langle p_2, 1 \rangle, \langle p_3, 2 \rangle\}$.

Example 3 Let us consider three students $\mathcal{D} = \{s_1, s_2, s_3\}$, and a measurement ψ_c which assigns each student to a class (foreign, local) identified with a number (1 = foreign, 2 = local): then $\psi_c = \{\langle s_1, 1 \rangle, \langle s_2, 2 \rangle, \langle s_3, 2 \rangle\}$.

It is important to remark that measurements are assignments (with further requirements). All the above ψ_t , ψ_q , and ψ_c are of course assignments; to be measurements they need to be “correct”, in order to satisfy the homomorphism requirement. For example, the assignment $\psi'_t = \{\langle o_1, 5 \rangle, \langle o_2, -5 \rangle, \langle o_3, 15 \rangle\}$ would not be a measurement when using Celsius degrees, assuming that the values in ψ_t are indeed correct.

Given a measurement, one can of course make some inferences from it and derive some further properties such as equalities, greater than / less than relationships, differences, ratios, ratios of differences, etc. These properties have been named “numerical statements” [4] or simply “statements” [2]. We use

²We assume that both empirical and numerical relational structures are pairs (a set with a single relationship). To be truly general, we should consider empirical and numerical relational structures having: (i) more than one relationship, (ii) relationships of different arity, and (iii) also functions besides relationships. This would mainly be a technical exercise and we refrain to do so to avoid unnecessary technical complications.

the latter term and we denote as ψ^* the set of statements that can be derived from the values assigned by ψ to the set of objects in $\mathcal{D}$. That is, $\psi^* = \{x \mid \psi \models x\}$. For instance, going back to our Example 1, ψ_t^* contains statements like: $\forall i \in \{2, 3\}(\psi_t(o_i) > 0)$ (“temperatures of o_2 and o_3 are above zero”); $\psi_t(o_3) > \psi_t(o_2) > \psi_t(o_1)$; $\psi_t(o_3) = 3 \cdot \psi_t(o_2)$; and $\frac{\psi_t(o_3) - \psi_t(o_2)}{\psi_t(o_2) - \psi_t(o_1)} = 1$.

Let us remark that such statements can be either “value oriented” (i.e., $\psi_t(o_3) = 15$) or “relationship oriented” (i.e., $\psi_t(o_3) = 3 \cdot \psi_t(o_2)$ or $\psi_t(o_2) < \psi_t(o_3)$). Provided that there are no errors, all the statements derived from a measurement are numerically correct. For instance, in our Example 1, $\psi_t(o_3) = -3 \cdot \psi_t(o_1)$ is numerically correct, while $\psi_t(o_3) = 4 \cdot \psi_t(o_1)$ is not. However, it makes no sense to assert that o_3 is minus three times hotter than o_2 ; whereas we can assert that there is the same difference of temperature between o_1 , o_2 and o_3 . That is, the statement $\frac{\psi_t(o_3) - \psi_t(o_2)}{\psi_t(o_2) - \psi_t(o_1)} = 1$ is numerically correct and makes sense, as well as the comparison $\psi_t(o_3) > \psi_t(o_2)$. In an analogous way, in the case of the Example 2, it makes no sense to state that there is the same difference in quality between the products p_1 , p_2 and p_3 although it is numerically true. We only know that some products have a higher quality than other products ($\psi_q(p_3) > \psi_q(p_2) > \psi_q(p_1)$). Finally, in the example of students and classes, there is no ordering relationship between classes: we can only say that two students belong to the same class or not.

According to the terminology used in measurement theory, we can summarize the above observations by saying that not all statements are *meaningful*. Meaningful statements are modeled in measurement theory on the basis of the concepts of *levels*, or *scales*.

2 Scales and Scale Types

The concepts of *scale* and *scale type* are central in measurement theory. The most typical categorization of scales is probably the one going from *Nominal* (N, the lowest scale type), through *Ordinal* (O) and *Interval* (I), to *Ratio* (R, the highest scale type). In our three examples above, temperature, product quality, and class assignment are I, O, and N scale types respectively. Other scales commonly used for temperature are of scale type I, like Fahrenheit, or R, like Kelvin. The choice of scale types is somehow arbitrary in the sense that there is not a single choice of which ones to define.

More formally, each scale type can be defined by means of an associated set of *permissible transformation functions*: each scale type corresponds to a particular set of functions.

Definition 3 (Permissible Transformation Functions) *The sets $\mathcal{F}_T$ of permissible transformation functions for each scale type $T \in \{N, O, I, R\}$ are (see also Table 1):*

- $\mathcal{F}_N = \{f \mid f \text{ is bijective, i.e., } x = y \text{ iff } f(x) = f(y)\};$
- $\mathcal{F}_O = \{f \mid f \text{ is strictly increasing monotonic, i.e., if } x < y \text{ then } f(x) < f(y)\};$
- $\mathcal{F}_I = \{f \mid f \text{ is a linear affinity, i.e., } f(x) = a \cdot x + b, a > 0\};$
- $\mathcal{F}_R = \{f \mid f \text{ is linear, i.e., } f(x) = a \cdot x, a > 0\}.$

Table 1: For each scale type, the set of permissible transformation functions $\mathcal{F}_T$ and the classic examples of Meaningful Statements

T	$\mathcal{F}_T$ (Permissible Transformation Functions)	$\widehat{\psi}_T^*$ (Meaningful Statements)
N	$\mathcal{F}_N = \{f \mid x = y \text{ iff } f(x) = f(y)\}$	$\widehat{\psi}_N^*(x, y) \iff x = y$
O	$\mathcal{F}_O = \{f \mid \text{if } x < y \text{ then } f(x) < f(y)\}$	$\widehat{\psi}_O^*(x, y) \iff x \geq y$
I	$\mathcal{F}_I = \{f \mid f(x) = a \cdot x + b, a > 0\}$	$\widehat{\psi}_I^*(x, y, z, w, k) \iff \frac{x-y}{z-w} = k$
R	$\mathcal{F}_R = \{f \mid f(x) = a \cdot x, a > 0\}$	$\widehat{\psi}_R^*(x, y, k) \iff \frac{x}{y} = k$

We also write $f(\psi)$:

$$f(\psi) = \{\phi \mid \phi \in \Psi, \phi(x) = f(\psi(x))\}.$$

Given these permissible transformation functions, the potential measurements over a set of objects are grouped into equivalence classes according to the following equivalence relation.

Definition 4 (Measurement Equivalence) Two measurements ψ and ψ' are equivalent for the scale type T if there exists a permissible transformation function $f \in \mathcal{F}_T$ such that ψ and $f(\psi')$ are identical:

$$\psi \approx_T \psi' \iff \exists f \in \mathcal{F}_T (\psi = f(\psi')).$$

The set of all equivalent measurements is the equivalence class $[\psi]_T$.

The basic idea is that permissible transformation functions determine if two measurements are equivalent or not. For the nominal scale type, measurements which distribute objects into the same groups are equivalent, given that there exists a bijective function which transforms values from one measurement to the other. For instance, the student distribution in Example 3 is equivalent to $\{\langle s_1, 4 \rangle, \langle s_2, 3 \rangle, \langle s_3, 3 \rangle\}$. The bijective function $f(x) = 5 - x$ transforms one measurement into another. For the ordinal scale type, measurements which order objects in the same way are equivalent. For instance, the measurement $\{\langle p_1, 4 \rangle, \langle p_2, 7 \rangle, \langle p_3, 10 \rangle\}$ is equivalent to the product quality measurement in Example 2. The monotonic transformation function which confirms that the two measurement are equivalent is $f(x) = 4 + 3 \cdot x$.

These examples illustrate two characteristics of measurement scale types. The first one is that they are additional features of measurements. That is, two measurements defined on different scale types could assign the same numbers to objects, and therefore, the derived statements are identical, but the set of *meaningful* statements would not be the same.

The second characteristic is that there exists a natural subsumption across measurement scale types. For instance, equality relations are meaningful in every scale type. Greater than and less than make sense in every scale type except in N. The difference between two values makes no sense in N nor in O. In other words, the more we consider “higher” scale types, the more we can state

meaningful statements. That is, the permissible transformation functions for high measurement scale types (e.g., ratio, or interval) are also transformation functions for low measurement scale types (e.g., ordinal or nominal):

$$\mathcal{F}_R \subset \mathcal{F}_I \subset \mathcal{F}_O \subset \mathcal{F}_N. \quad (1)$$

Therefore, there exists also a subsumption between equivalence of measurements across scale types. That is, two equivalent measurements for a high scale type are also equivalent for lower scale types:

$$\psi \approx_R \psi' \implies \psi \approx_I \psi' \implies \psi \approx_O \psi' \implies \psi \approx_N \psi'.$$

We denote the ordering relationship on the set of the four scale types as

$$N < O < I < R, \quad (2)$$

and we speak of higher and lower scale types accordingly.

We briefly mention a property that is further discussed in the rest of the paper: there is a dependence between tasks and scale types. Roughly, classification and clustering are based on the nominal scale type; ranking on the ordinal, and quantitation on the interval and ratio. However this dependence is not clear cut, as we discuss in more detail in the paper.

3 From Assignment to Measurements and Back

Let us remark that although measurement theory is focused on measurements, some parts of it can be immediately generalized to assignments. More specifically, the notions of equivalent measurements and of equivalence class $[\psi]_T$ of a measurement ψ of Definition 4 can be generalized to assignments in a straightforward way. Thus, given an assignment, it is possible to speak of an *equivalent assignment for a given scale type T*. The following definition formalizes the simple generalization of Definition 4.

Definition 5 (Assignment Equivalence) *Two assignments ω and ω' are equivalent for the scale type T if there exists a permissible transformation function $f \in \mathcal{F}_T$ such that ω and $f(\omega')$ are identical:*

$$\omega \approx_T \omega' \iff \exists f \in \mathcal{F}_T (\omega = f(\omega')).$$

The set of all equivalent assignments is the equivalence class $[\omega]_T$.

The reason behind this definition is that if an assignment is not a measurement, then it means that there is no homomorphism between the empirical and numerical relational spaces, i.e., the assignment values are not correct w.r.t. the “real situation” (since measurement theory is interested in correct measurements, these cases are usually not taken into account); but in Definitions 4 and 5 there is no reference to the homomorphism.

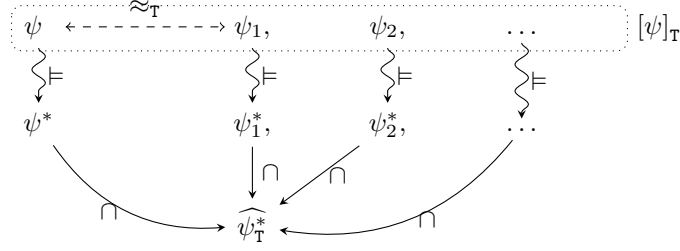


Figure 1: Diagram of equivalent measurements (dashed line labelled with $\approx_T$), equivalence classes (dotted rectangle labeled with $[\psi]_T$), derivation (zigzag lines labelled with $\models$) of statements (ψ^*), and intersection (continuous lines labeled with $\cap$) to get the meaningful statements ($\widehat{\psi}_T^*$).

4 Meaningfulness

We can now address the last element of measurement theory of interest here, namely the so called *meaningfulness problem* [4]: as we have seen, given a measurement ψ , not all the statements in ψ^* (sometimes not even some of those in ψ) are meaningful.

Commonly, measurement scale types are described in terms of what statements can be stated over two or more objects: the nominal scale type focusses on the equality relationships, the ordinal scale type incorporates the greater than and less than relationships between values, the interval scale type adds ratios of differences, and the ratio scale type includes ratios. Suppes and Zinnes [4] define a “meaningful numerical statement” as a numerical statement which is constant under permissible transformation functions, that is, under an equivalent measurement. In this paper, we refer to this as *Meaningful Statement*.

Definition 6 (Meaningful Statement) *Given a measurement ψ of scale type T , the set of its Meaningful Statements, denoted as $\widehat{\psi}_T^*$, contains the statements that are invariant across equivalent measurements:*

$$\widehat{\psi}_T^* = \{x \in \psi^* \mid \forall \psi_i \approx_T \psi (x \in \psi_i^*)\} = \bigcap_{\forall \psi_i \approx_T \psi} \psi_i^*.$$

Figure 1 shows how *equivalence* and *meaningfulness* are related to each other. At the top left of the figure, there is the measurement ψ . On its right there are its equivalent measurements (ψ_i) on the same domain and for the same scale type T . All of them belong to the equivalence class $[\psi]_T$. For instance, they could be different measurement of the ordinal scale type which keep the same order across object values. Each measurement produces different derived statements (ψ^*, ψ_i^*), but the meaningful statements for the scale type T are those that are shared by every derived measurement.

According to this definition, we can infer certain meaningful statements for each scale type. Figure 2 shows a graphical representation of some meaningful

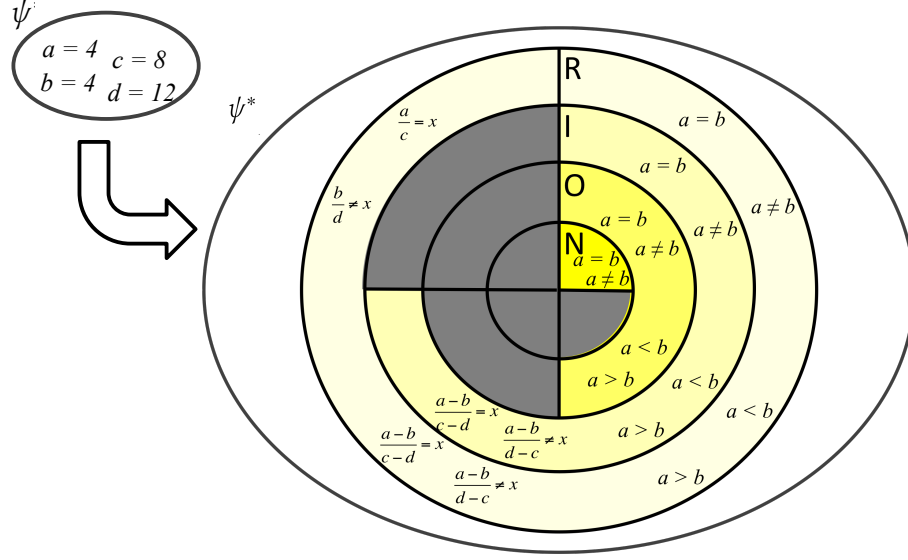


Figure 2: (Meaningful) statements and scale types.

statements for each scale type. As the figure shows, there is a subsumption across scale types, being the equality the most basic relationship in the nominal scale type. It is important to remark that those listed in the figure are not all the possible meaningful statements. For example, one could compute the arithmetic mean of some temperatures in Example 1 as $\text{mean}(\psi_t(o_1), \psi_t(o_2)) = 0$, $\text{mean}(\psi_t(o_2), \psi_t(o_3)) = 10$ and state that the difference between the two means is the same as $\psi_t(o_2) - \psi_t(o_1)$, and this would be true under any permissible transformation function in $\mathcal{F}_I$, i.e., meaningful.

The most important aspect of this figure is the grey area that highlights how some statements for certain scale types are not meaningful. For instance, the ordinal relationship between values ($a > b$) is not invariant across equivalent measurements of the nominal scale type. Therefore it is hidden by the grey area.

As shown in the last column of Table 1, equality is a boolean 2-ary meaningful relationships for the N scale type; greater than ($\geq$) is a meaning relationship for the O scale type; the ratio of differences is a 5-ary meaningful relationship for the I scale type; and the ratio is a 3-ary meaningful relationship for the R scale type.

It is also important to remark that, perhaps surprisingly but obviously, for any measurement ψ , the assignments of single values in ψ (e.g., $\psi(d_1) = 6$) are *never* meaningful.³ none of them is invariant under the corresponding permissi-

³The only case for which this would be different is the sometimes proposed *absolute* scale

ble transformation functions. In other terms, to find something meaningful one needs to look inside the set of derived numerical statements ψ^* . For example, $\psi_t(d_1) = -5$ of Example 1 is false when changing to Fahrenheit, although “the real temperatures are always the same”. This statement captures something that is not “in the empirical relational structure” [4], and does not represent properties of the “true” temperatures; rather, it is an artefact of a particular measurement approach.

It is also important to understand that meaningfulness is not always an important, or even desired, property. If one is given the task “measure temperature using the Celsius scale”, then providing an equivalent measurement in Fahrenheit is not a satisfactory achievement of the task. Also this issue is discussed in the paper.

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type, for which the only permissible transformation function is identity.